CivicPulse - Concept Revision Bed File
Code Yaad Nahi, Kahani Yaad Rakho - Viva 10 Min Me!

PILLAR 1: AUTH - "Chor ko Admin nahi banne dena!" **Kahani: Hacker Postman se {role:"admin"} bhej ke admin ban jata! **Kiya:** Register pe role body se lena band! Hardcoded citizen! OTP generate + DB INSERT + 10min expiry + is_used=FALSE! Login pe 3 checks: isVerified? isActive? password? JWT 15m Access + 7d Refresh + httpOnly + Rotation (har bar naya token, purana delete)! Forgot me same message "If email registered..." -> Enumeration block! **Viva Q:** Role hardcoded kyun? A: "Sir, Privilege Escalation rokne ke liye!" **Viva Q:** Rotation kya? A: "Sir, token theft se bachao! Har refresh pe naya token, purana invalidate!"**

PILLAR 2: DUPLICATE + GIS - "100m me ek hi report!" **Kahani: Same gully me 10 log same pothole report - Waste! **Kiya:** Haversine Formula lat/lng se distance km! 0.1km=100m andar same category to 409 Conflict + "Existing ko upvote karo"! Area auto-detect ST_Contains(boundaryCoords, POINT(lng lat)) - Karachi 4 polygons seed me! location_point=POINT(lng lat) + lat/lng columns - BBOX fast! **Viva Q:** 100m check kyun? A: "Sir, duplicate rokne + upvote se priority badhe!" **Viva Q:** Polygon kyun? A: "Sir, point kis area me auto pata chale!"**

PILLAR 3: SLA SCHEDULER - "Har 15 min Chowkidar!" **Kahani: Deadline cross hua to kaun batayega? **Kiya:** cron.schedule("*/15 * * * *") - Har 15 min PDF! WHERE deadline < NOW() AND slaBreached=FALSE! FOR UPDATE Lock -> 2 server same row double notify na kare! Race condition block! Priority: severity*10 + upvoteCount + HOUR/6 - Purana auto upar! Manager Dynamic JOIN departments + manager_id - Hardcode 1 nahi! Transaction BEGIN+COMMIT/ROLLBACK + stopSLAScheduler() on SIGTERM! **Viva Q:** FOR UPDATE kyun? A: "Sir, race condition - Duplicate notification rokne!" **Viva Q:** Priority kaise badhti? A: "Sir, Severity*10 + Upvotes + Har 6 ghante +1! Cron har 15 min recalc!"**

PILLAR 4: REAL-TIME - "Colon wala Room!" **Kahani: Pehle user_123 underscore tha, controller me user:123 colon - Notification jati hi nahi thi! **Kiya:** Socket Handshake Auth io.use() + handshake.auth.token + jwt.verify - Bina login koi connect nahi! Rooms: user:${userId} personal -> Sirf us user ko! issue:${issueId} group -> Live tracking! getIO() global getter -> Controllers se io.to(user:123).emit()! Fix Colon format user: Privacy safe! **Viva Q:** Room kyun? A: "Sir, privacy! io.emit() sabko jayega - Data leak! Room se sirf usi user ko!" **Viva Q:** Handshake auth kyun? A: "Sir, bina login events sun sakta hai - JWT mandatory PDF!"**

PILLAR 5: SECURITY - "Secret GitHub pe nahi!" **Kahani: JWT_SECRET || "supersecretkey" fallback - GitHub pe sabko secret! PDF AUTOMATIC FAIL! **Kiya:** if (!SECRET) throw Error("Must be set in .env!") - Fail fast! Server start hi nahi hoga! httpOnly:true -> JS document.cookie se nahi padh sakte - XSS block! sameSite:strict -> Cross site cookie nahi - CSRF block! secure: production -> Prod HTTPS only! Ownership Check WHERE id=? AND user_id=? + affectedRows===0 =>404 -> IDOR protection! State Machine VALID_STATUS_TRANSITIONS Single Source -> Submitted->Resolved direct 409! Rate Limiter authLimiter + apiLimiter + limit max 100 -> DoS block! CSV Escaping " -> "" + "..." wrap! Image Sharp 1200px + WebP 80% + thumb 300px + withoutEnlargement -> 5MB->200KB! **Viva Q:** Hardcoded fallback kyun hatao? A: "Sir, secret committed = automatic fail! Throw error fail fast!" **Viva Q:** IDOR kya? A: "Sir, User 1 user 2 ka issue ID guess karke edit! AND user_id=? se rokta!"**

VIVA KE 15 SAWAL - 2 LINE JAWAB! 1. OTP Fix? "Sir, OTP generate par DB INSERT missing tha - Verify fail! INSERT add kiya!" 2. JWT 15m+7d kyun? "Sir, Access short Security! Refresh long + Rotation + httpOnly!" 3. Bcrypt 10 rounds? "Sir, PDF requirement! Hash slow taaki brute force slow!" 4. Haversine? "Sir, lat/lng se distance km! Duplicate 100m check!" 5. ST_Contains? "Sir, GIS - Point polygon andar hai ya nahi!" 6. Median vs AVG? "Sir, AVG me ek 100 din wala AVG bigad deta! Median beech wala Real!" 7. Window Function? "Sir, ROW_NUMBER() OVER(PARTITION BY area) - Har area ka beech wala! MySQL 8 Topper code!" 8. Sharp kyun? "Sir, C++ libvips - Jimp se 10x fast! WebP 30% chhota!" 9. withoutEnlargement? "Sir, chhoti image 200px ko 1200px bada to blur! Isliye!" 10. Transaction? "Sir, 2 queries - Ek fail to dono rollback! Data corrupt nahi! Verify me isVerified + is_used ek saath!" 11. INSERT IGNORE seed? "Sir, Re-runnable! Dobara npm run seed to duplicate error nahi! Demo!" 12. Swagger? "Sir, PDF API docs! Frontend bina Postman test kare!" 13. Coverage 70%? "Sir, PDF mandatory! Jest threshold 70% - Kam to npm test fail!" 14. Graceful Shutdown? "Sir, SIGTERM pe server band - Pehle cron stop, phir DB pool end - Duplicate cron nahi!" 15. TicketNo? "Sir, CP-2026-000123 - Readable + Year + Sequence! MAX(id)+1 se!"

DEMO FLOW - VIVA ME YAHI DIKHANA! (Code Nahi!) Step 1: Seed npm run seed -> 4 Areas + 4 Depts + 5 Cats + 4 Users (admin/manager/officer/citizen) + 3 Issues Ready! Logins print! Step 2: Citizen Login citizen@civicpulse.com / Password123! -> isVerified=TRUE direct! Step 3: Report Issue Lat 24.85 Lng 67.05 Category Pothole -> Area auto-detect Downtown North! TicketNo CP-2026-...! Step 4: Duplicate Check Same lat/lng + same category dobara -> 409 "Already reported within 100m - Upvote existing!" Step 5: Manager Login

manager@civicpulse.com -> Dashboard: total, open, resolved, SLA breach, Officer workload! Step 6: SLA Breach Demo Console [SLA Alert] 1 new issue breached! -> Citizen + Manager notification + Socket user:3 room! Step 7: Officer Resolve Officer login -> Resolve with proof photo mandatory -> WebP compress -> Status InProgress->Resolved Step 8: Public Map /api/v1/issues/map?lat=24.85&lng;=67.05&radius;=1 -> BBOX + Radius + LIMIT 500! Bas! 5 min Full Backend Demo!

SONE SE PEHLE 10 MIN REVISION 1. 5 Pillars ki Kahani: Chor-Admin, 100m, Chowkidar, Colon Room, Secret GitHub! 2. 5 Viva Answers: FOR UPDATE, Rotation, Haversine, IDOR, Hardcode Fail! 3. Demo Flow: Seed -> Citizen Report -> 409 Duplicate -> Manager Dashboard -> SLA Alert -> Officer Resolve! Code Yaad Nahi, Concept Yaad Hai - Aap Topper Ho! Good Night!